



KE: Ontology Overview

Hutchatai Chanlekha

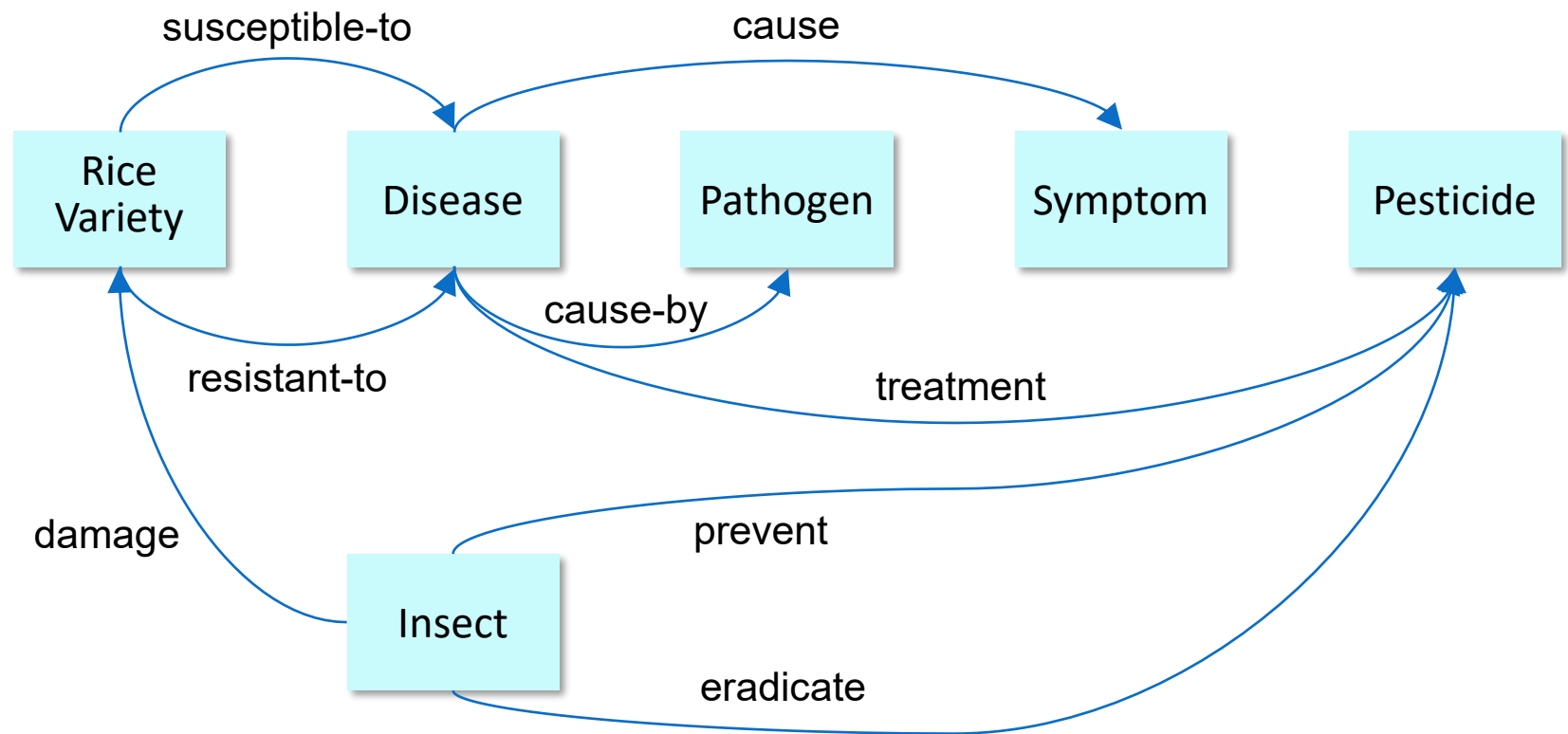
Department of Computer Engineering, Kasetsart University





Suppose we want to model the knowledge about rice diseases

Let's think about important objects and their relations in the domain



Suppose we want to incorporate knowledge about insects that may harm rice crops.



We want to ...

- Model knowledge as main domain concepts, their relations and properties, along with axioms and restrictions
- In a way that makes it easy to incorporate additional knowledge
- In a way that allows knowledge to be shared and reused



What is Ontology?

- In Philosophy
 - To understand/define the nature of being, becoming, existence or reality as well as the basic categories of being and their relations
- In Information Science or Computer Science
 - To define formal definition of the types, properties, and interrelationships of the entities
 - In a way that is understandable for both human and computer
- Common in both sciences
 - Study of **concepts**, their **property** and **relation**



Ontology and Semantic Web

- Ontology is one of the building blocks of Semantic Web technology.
- An ontology is a formal description of knowledge as a set of concepts within a domain and the relationships that hold between them.
 - It includes machine-interpretable definitions of basic concepts in the domain and relations among them.
- To enable such a description, we need to formally specify components, such as individuals (instances of objects), classes, attributes and relations as well as restrictions, rules and axioms.



Components of Ontology

- An ontology is a formal explicit description of
 - **Concepts** (or classes) in a domain of discourse,
 - **Properties** (or roles or facets) of each concept describing various features and attributes of the concept
 - **Restrictions** on the properties (sometimes called property restrictions), including **rules** and **axioms**.
- An ontology together with a set of **individual instances** of classes constitutes a knowledge base.



Elements in ontology: Classes

- Classes are the focus of most ontologies.
- Ontology use classes to describe concepts in the domain.
 - For example, a class of books represents all entities that are books. Specific books are instances of this class.
- A class can have subclasses that represent concepts that are more specific than the superclass.
 - For example, we can divide the class of all wines into red, white, and rosé wines.
 - Alternatively, we can divide a class of all wines into sparkling and non-sparkling wines.

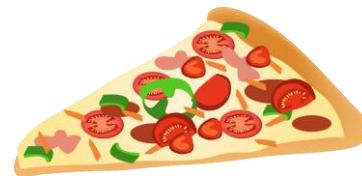
What is Concept?

- An aspect of things or abstract entities in one's mind
- Referring concepts with words
 - Label
 - Definition

Concept : “Food”

Label : food, อาหาร, 食物, 食べ物

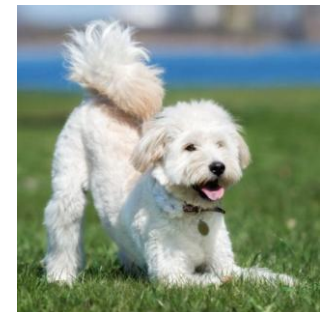
Definition : any nutritious substance that beings eat or drink in order to maintain life and growth



Concept (class) VS Individual (instance)

- Domesticated carnivorous mammals that typically has a long snout, sharp teeth, and a coat of hair.
- They have four legs ending in paws with toes and nonretractable claws.
- They are known for their sharp senses, particularly their sense of smell, and their ability to bark, howl, or whine.
- They are typically kept as a pet or used for various purposes like hunting, herding, or guarding.

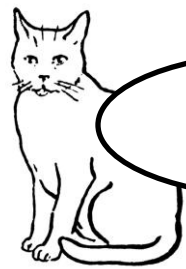
Dog Concept



Dog Individual

Concepts VS Instances

- Instance : single occurrence/ individual of concept
- Instance must have identity

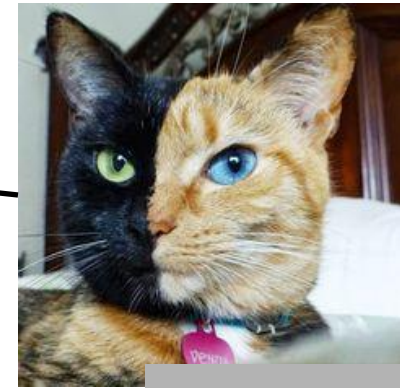


Cat

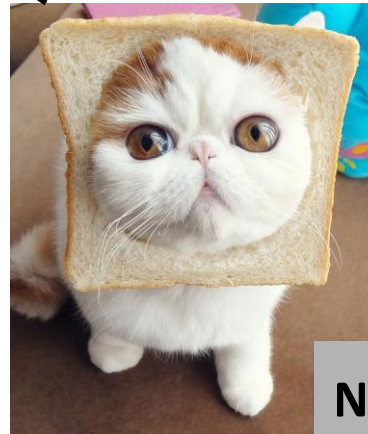
Instance of

Instance of

a small domesticated carnivorous mammal with soft fur, a short snout, and retractable claws. It is widely kept as a pet or for catching mice, and many breeds have been developed.



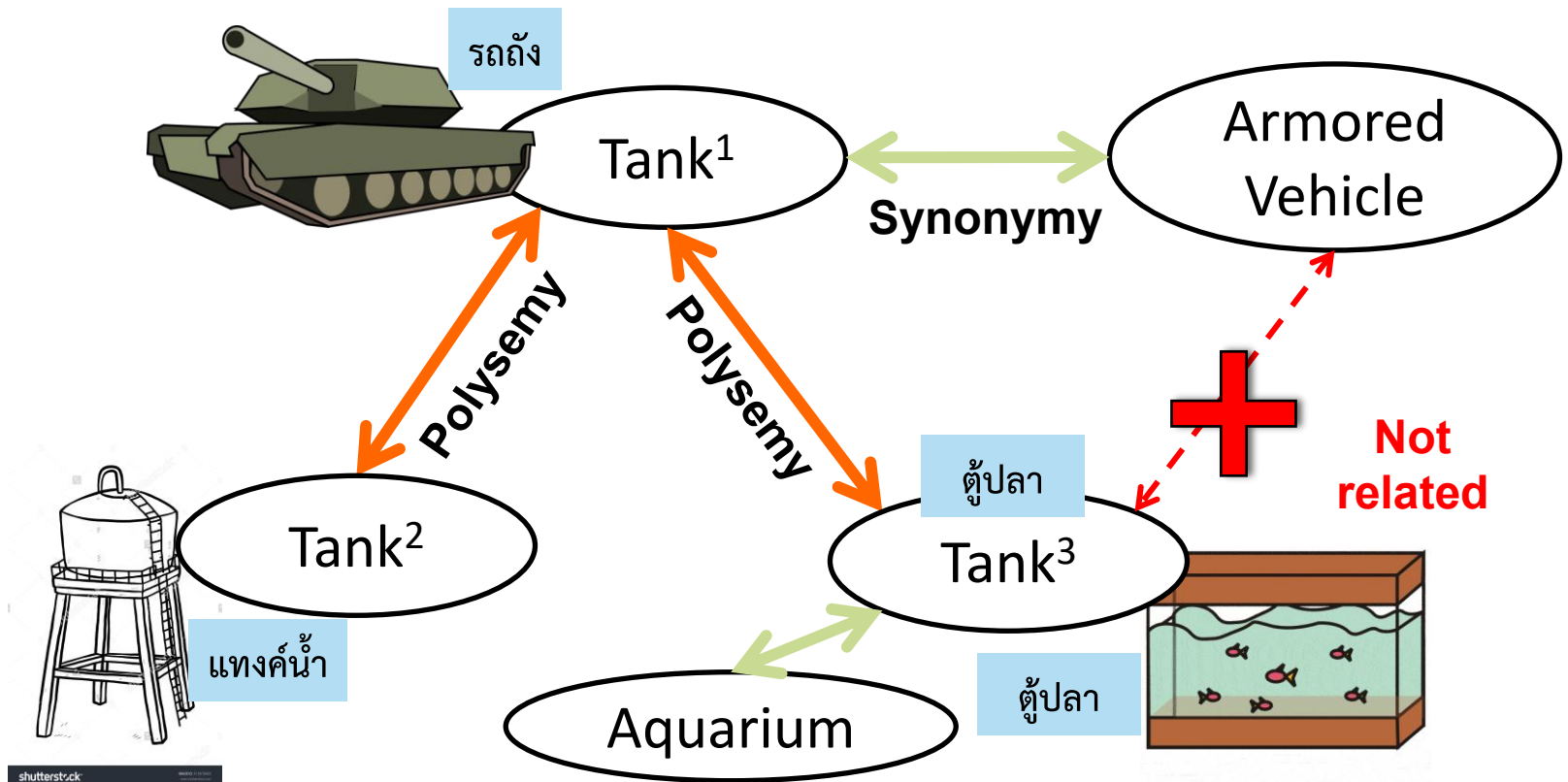
Name : Venus
Gender : Female



Name : Snoopy
Gender : Female

Concepts VS Labels (controlled vocab)

- Every legit terms refer to a concept
- Concepts may have many labels (synonymy; language-dependent)
- Same Labels may refer to various concepts (polysemy, homonym)





Although same label can be used to refer to various concepts,

Don't use same labels to represent different concepts
in an **ontology**



OWL ontology: Class VS Individual

- In building ontologies, class and individual distinction is frequently blurred:
 - ***Class vs. Individual (Instance)***
 - In certain contexts, something that is obviously a class can itself be considered an instance of something else.
 - For example, in a Dog ontology, *Golden Retriever*
 - Could it be an individual that is an instance of Dog? OR a subclass of Dog?
 - *GoldenRetriever* could be considered a class, the set of all actual Golden Retriever dogs.
 - For example, application for Veterinary hospitals
 - *GoldenRetriever* is an instance of class Dog, denoting the actual dog breed called Golden Retriever.
 - For example, application for dog breed suggestion
 - Note that the development of an ontology should be firmly driven by the intended usage.



Class hierarchy

- In ontology, classes can be organized into hierarchy via is-a relation.
- IS-A relation
 - Superclass-Subclass or Parent-Child relation
 - A superclass is more general than its subclass.
 - A is a B means B is a superclass of A
 - Anything that belongs to a subclass also belongs to its superclass.

- For example:

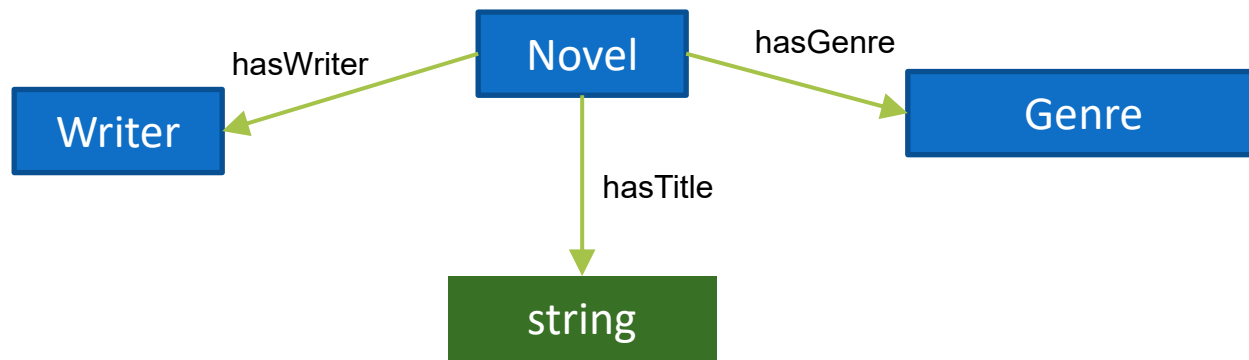
Superclass	Subclass
Fruit	Orange, Durian, Mango
Mammal	Dog, Cat, Cow
Food	Pizza, Sukiyaki, Noodle
Vehicle	Car, Motorcycle, Boat

- Subclasses must inherit properties of their superclass!



Elements in ontology: Properties

- Ontology provides mechanism to describe properties of classes and instances.
- Property : Relations to other concepts and the data attribute that define the concept
 - For example: Novel
 - Properties of book novel could be, such as, title, writer, genre, publisher, etc.

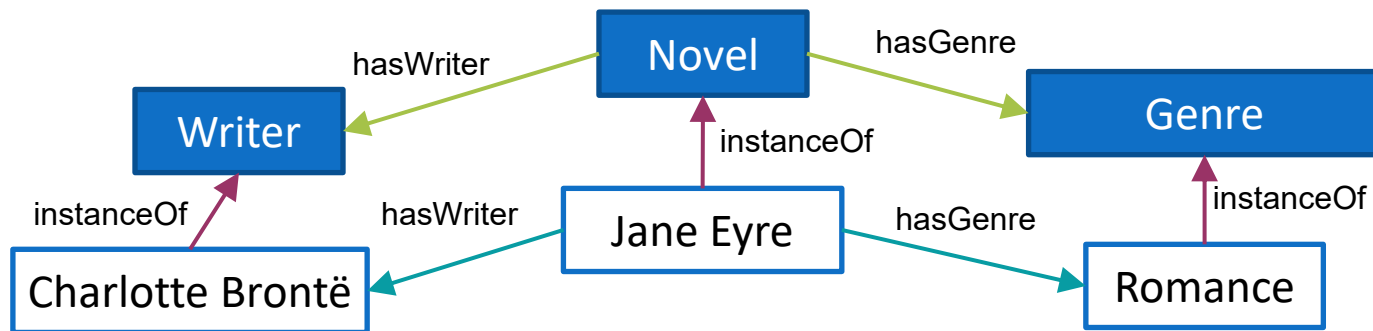




Elements in ontology: Properties

For example: Novel “Jane Eyre”

- The author of the novel “Jane Eyre” is Charlotte Brontë; it’s genre is romance novel.
 - In this example: at class level, we have class “**Novel**”, “**Writer**”, “**Genre**”
 - We can have properties describing the “Novel”, such as “has_writer” and “has_genre”

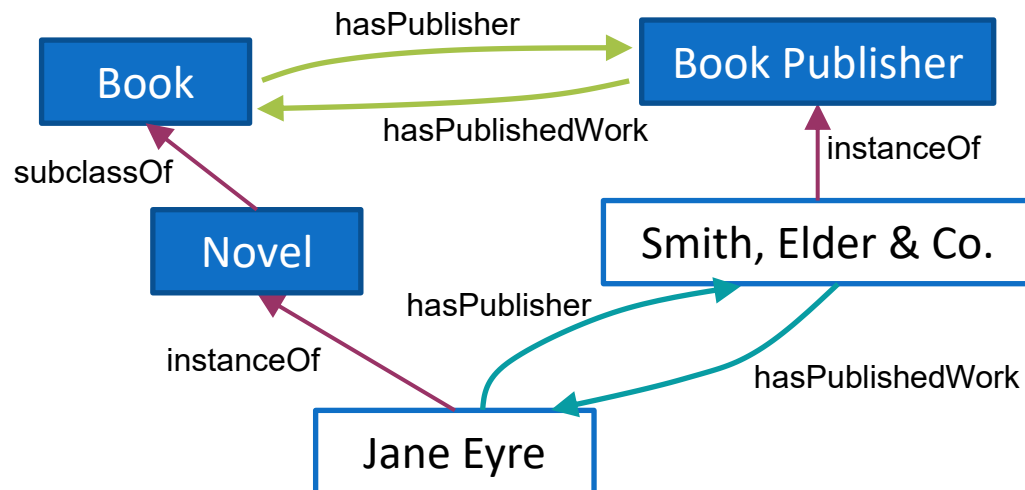


- We can also define other properties for class Novel describing their title, subtitle, publisher, original publication date, copyright status, language of work, and so on.



Elements in ontology: Properties

- Properties can be inherited from a superclass to its subclasses.
- For example:
 - Suppose the class 'Book' has the property 'has_publisher'.
 - This means that not only instances of 'Book', but all of its subclasses and their instances will also have the 'has_publisher' property.
 - Suppose the class 'Book Publisher' has a property 'has_PublishedWork' that refers to all the books that the book publisher published.
 - This will also refer to all instances of 'Book' and its subclasses





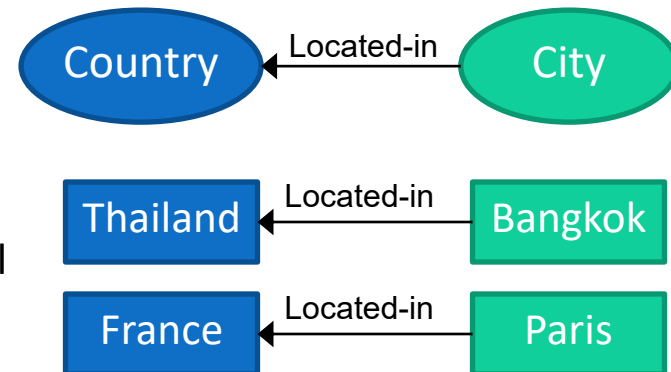
Constraint on property

- We can define constraint on a property
- For example: cardinality restriction
 - Maximum cardinality
 - What is the maximum number of a biological mother that a person can have?
 - Minimum cardinality
 - What is the minimum number of children a person can have?
 - Exact cardinality
 - How many wheel that a car can exactly have?

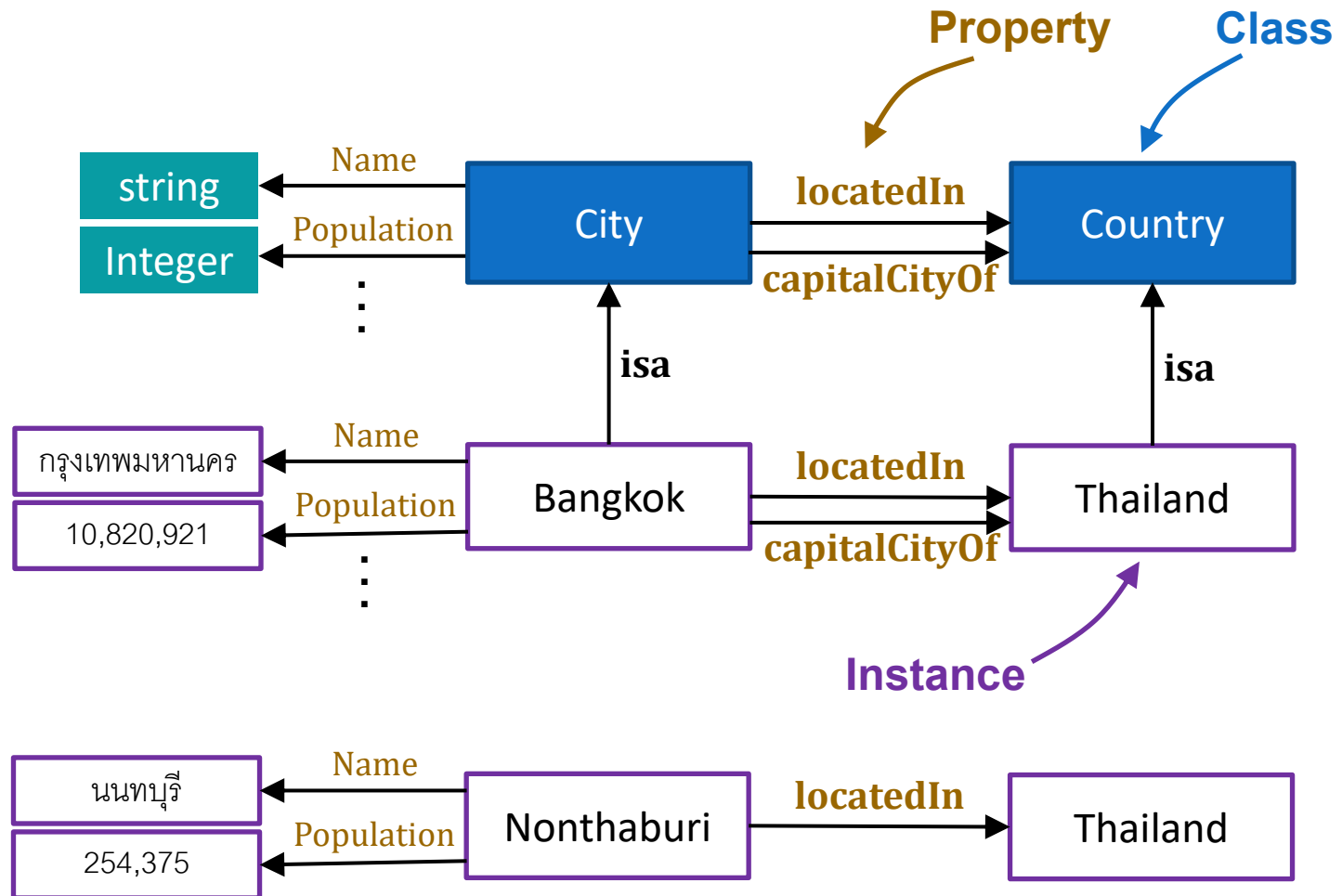


Ontology components

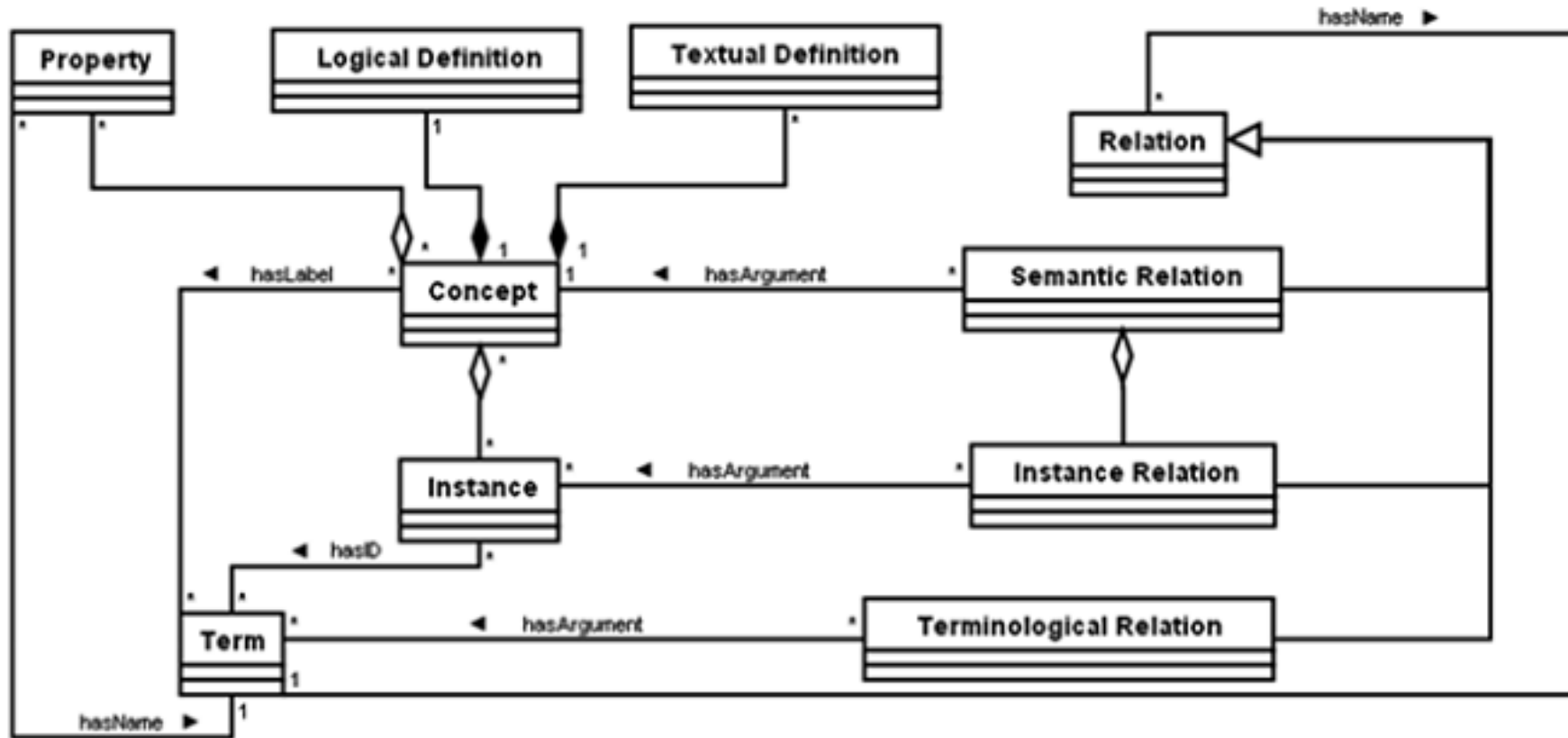
- Ontology components: **Concepts, properties, and instances**
 - Referenced by one or more symbols (or labels)
 - Symbols are terms that humans can understand roughly by reading them
- Concepts (and instances) are connected through relations/properties
 - Semantic relations: link only concepts together
 - Instance relations: connect only instances
 - Instance of semantic relations
 - Relations between instances can be contextual and cannot be generalized to all instances of their concept
 - Example of contextual instance relation: A city can have a “capital_city_of” relation with country. But only Bangkok that has a “capital_city_of” relation with Thailand.
- Terminological relations express the relationships that terms can have
 - Example: term “person” is synonym to the term “human being”
term “สุนัข” is a label of a class “Dog”



Components of Ontology



Components in Ontology



Class 1 —◇— Class 2

Aggregation association:
Objects of class 2 aggregate
objects of class 1

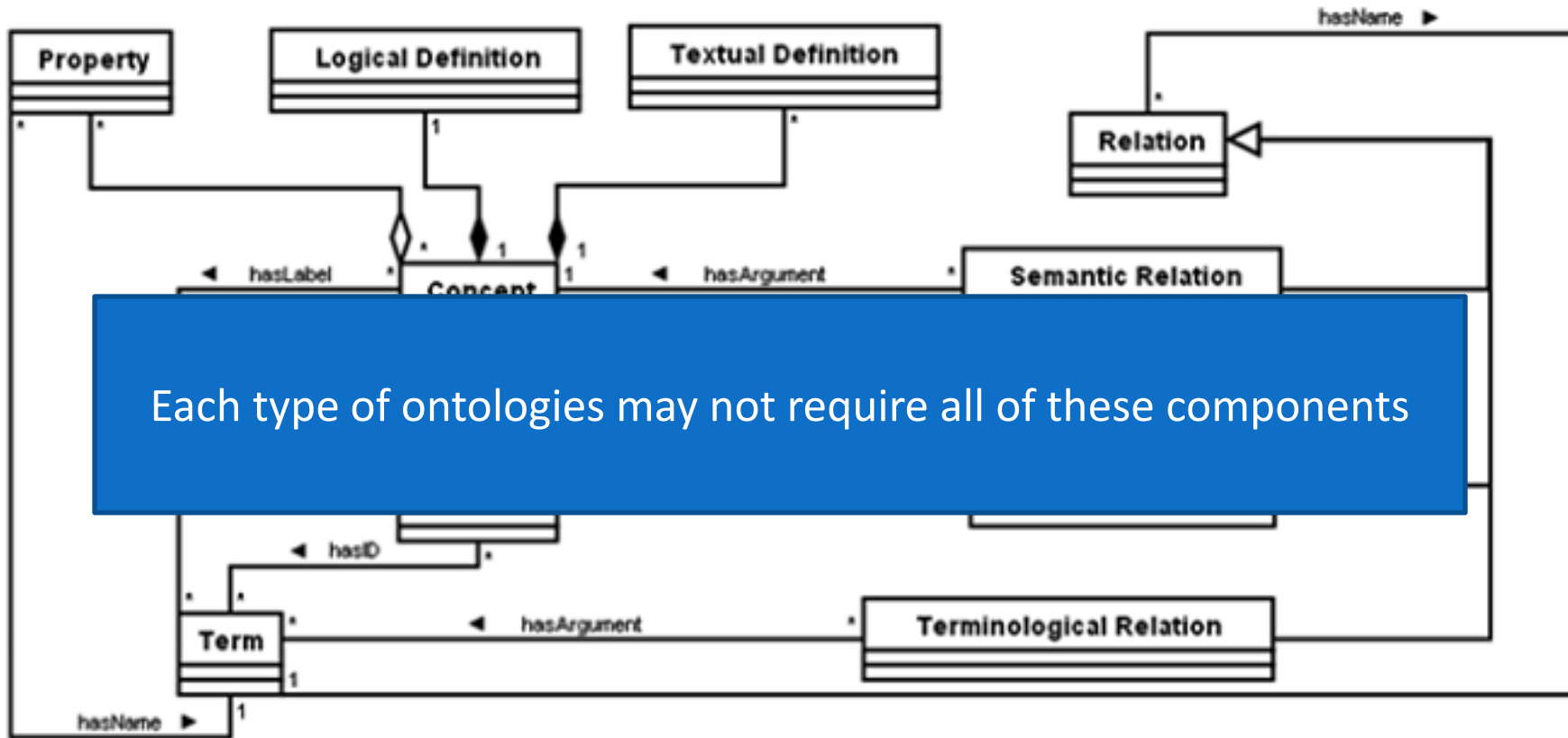
Class 1 —◆— Class 2

Composition association: Objects of class 2 is
composed of objects of class 1; the destruction
of objects of class 2 implies the destruction of
associated objects of class 1

Class 1 —▷— Class 2

Generalization relationship:
Objects of class 2 generalizes
objects of class 1

Components in Ontology



Each type of ontologies may not require all of these components

Class 2 —◆— Class 1

Aggregation association:
Objects of class 2 aggregate
objects of class 1

Class 2 —◆— Class 1

Composition association: Objects of class 2 is
composed of objects of class 1; the destruction
of objects of class 2 implies the destruction of
associated objects of class 1

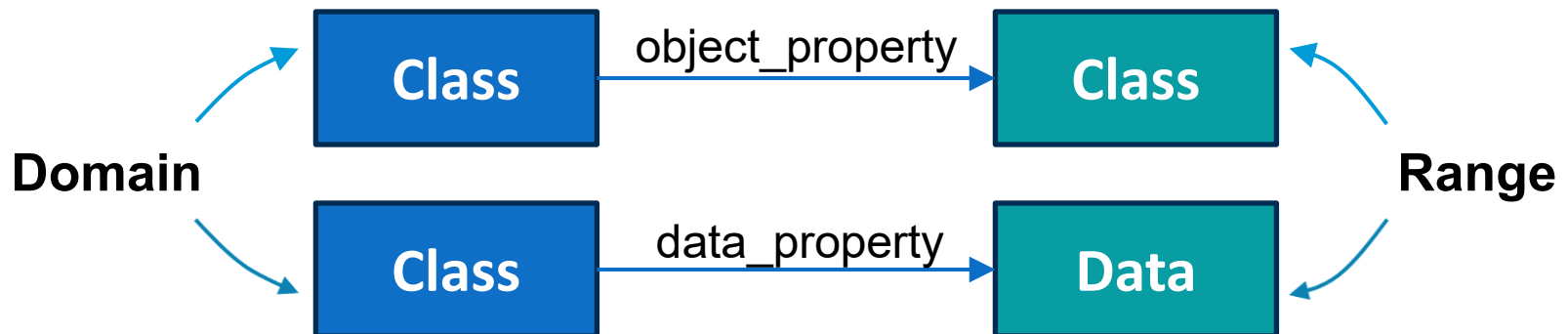
Class 2 —▷— Class 1

Generalization relationship:
Objects of class 2 generalizes
objects of class 1



Remark

- For Protégé
 - Relations between objects (either class level or individual level) are called **Object Property**
 - *object properties*: relate classes to classes (or individuals to individuals)
 - Properties of objects whose values are predefined data type are called **Data Property**
 - *data properties*: classes to datatypes (or individuals to datatypes)



IS-A

Part-of (object property)

Attribute-of (data property)

Bird



Bedroom



Expiration Date



IS-A

Part-of (object property)

Attribute-of (data property)

Hand

Fish

Sport

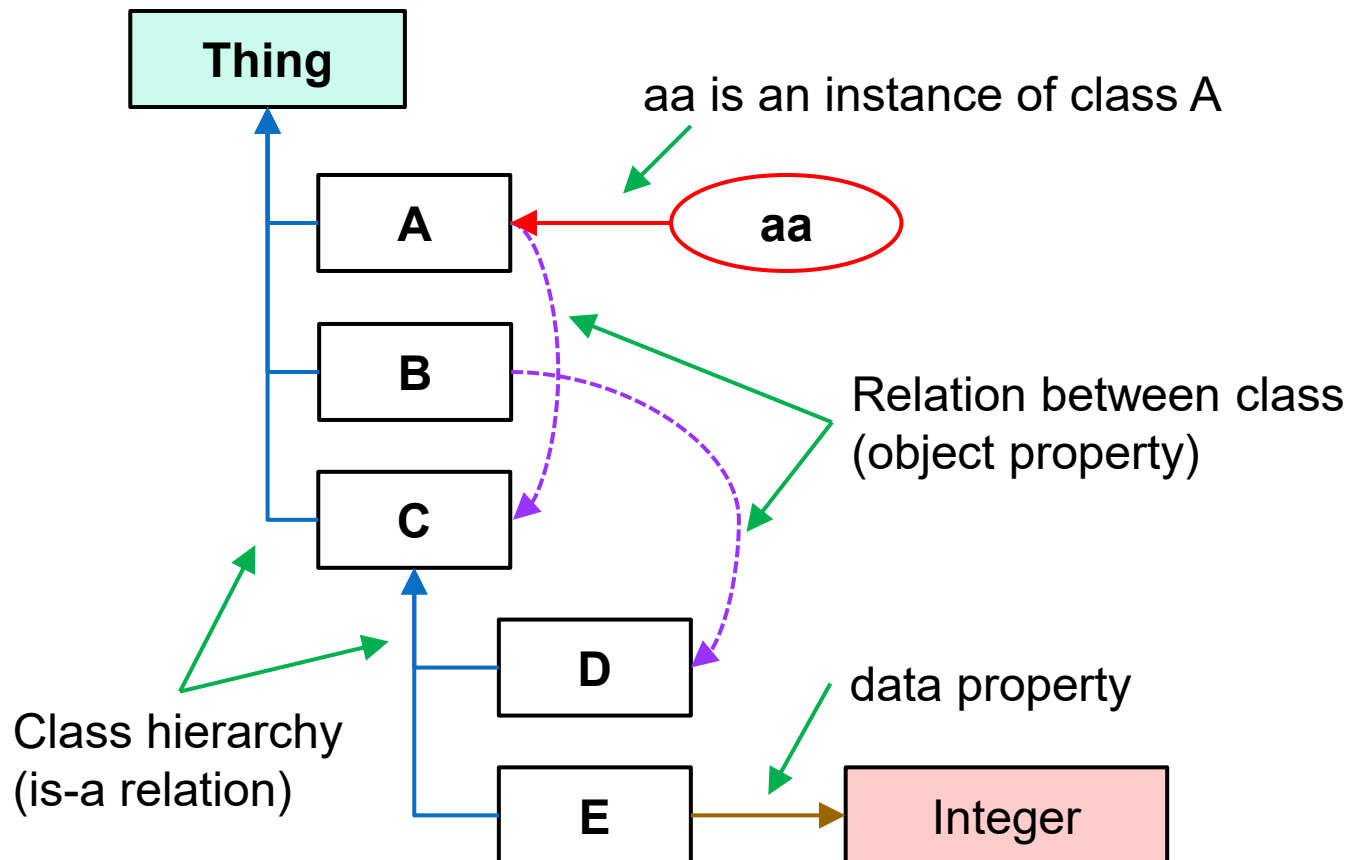
Let's test it out

Nail



Designing Ontology Exercise

- Use below notation to represent class hierarchy, class instance, and has-part relationship

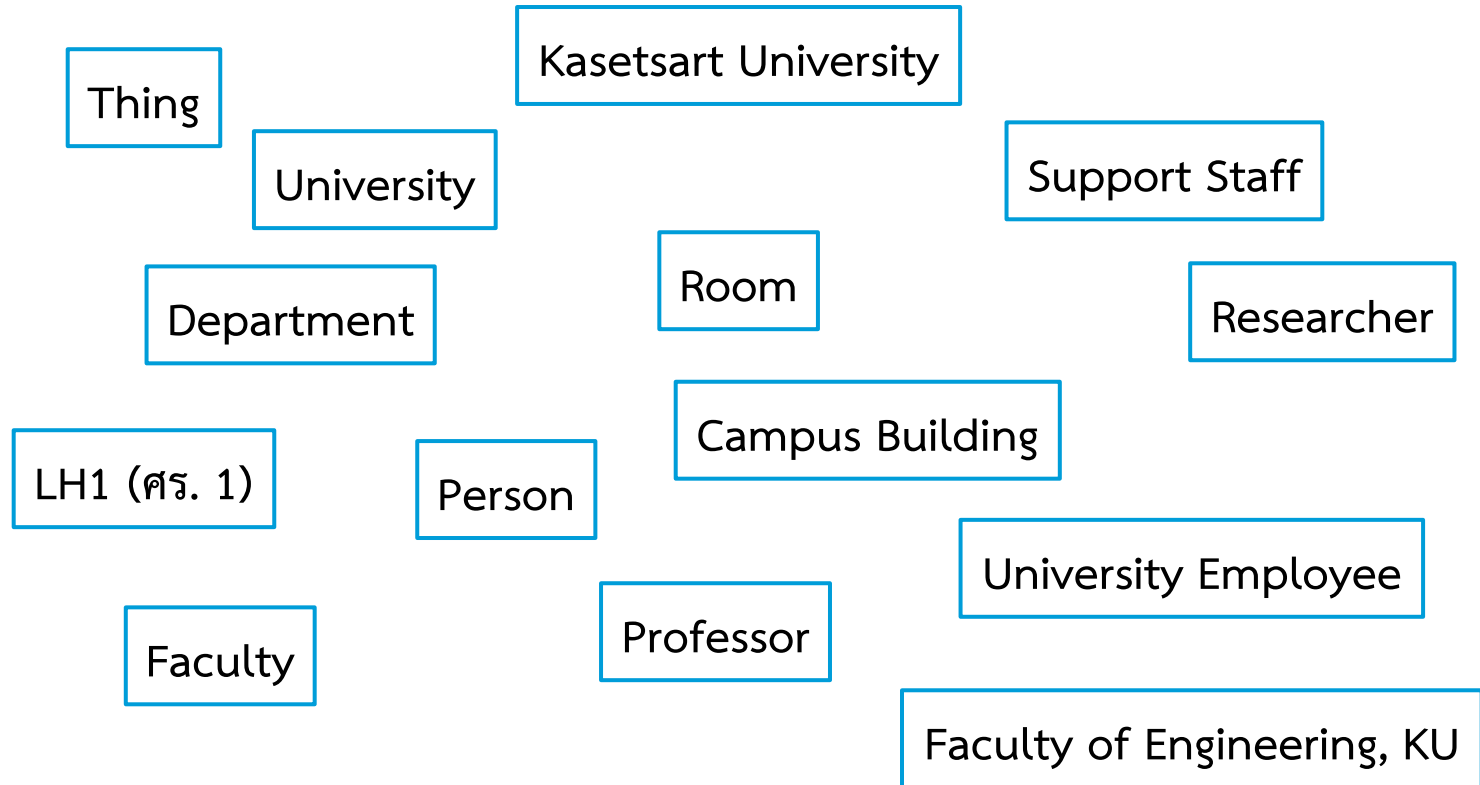




Exercise

From the list of items below, create an ontology about the university by arranging them into a class hierarchy, as well as add relation (object property) and instance-of relations where appropriate.

(object property relation includes A has B, A has B as its member, A has B in it, A owns B.)

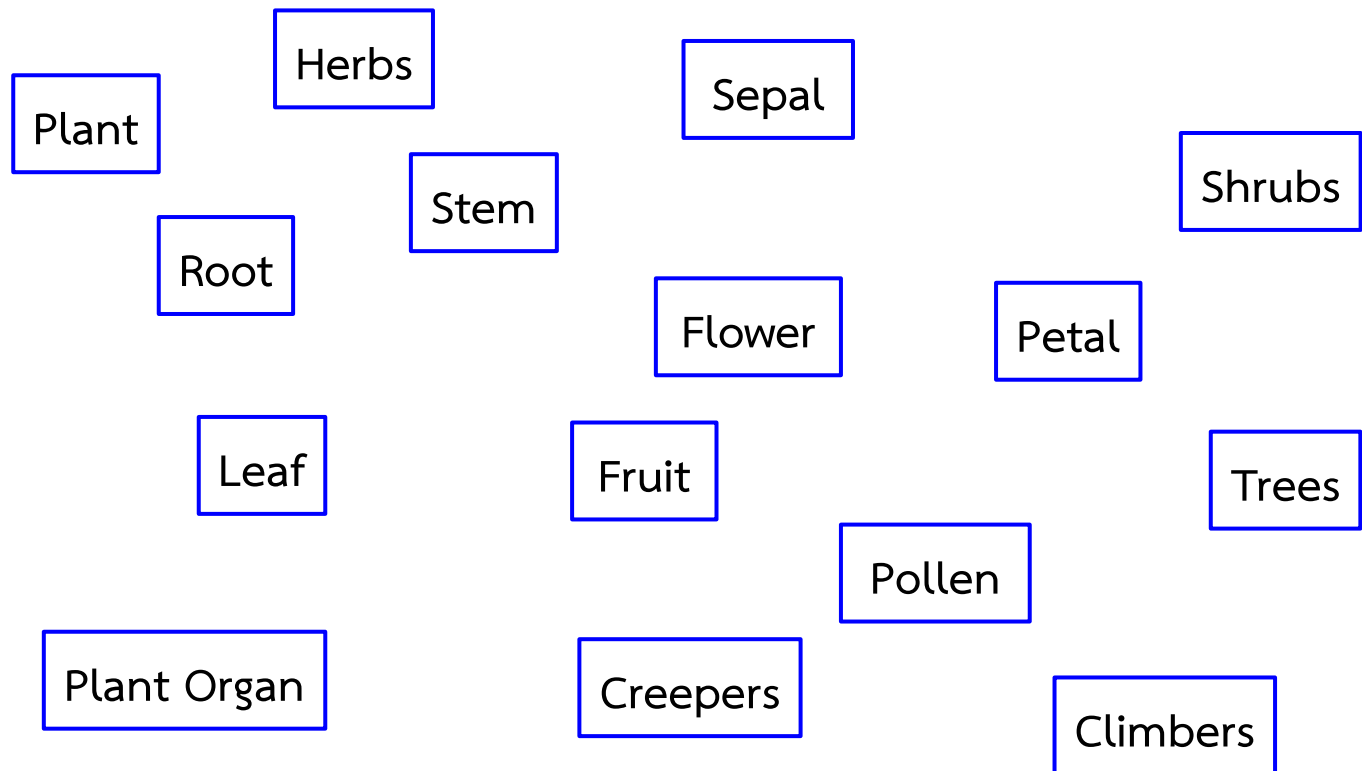




Exercise

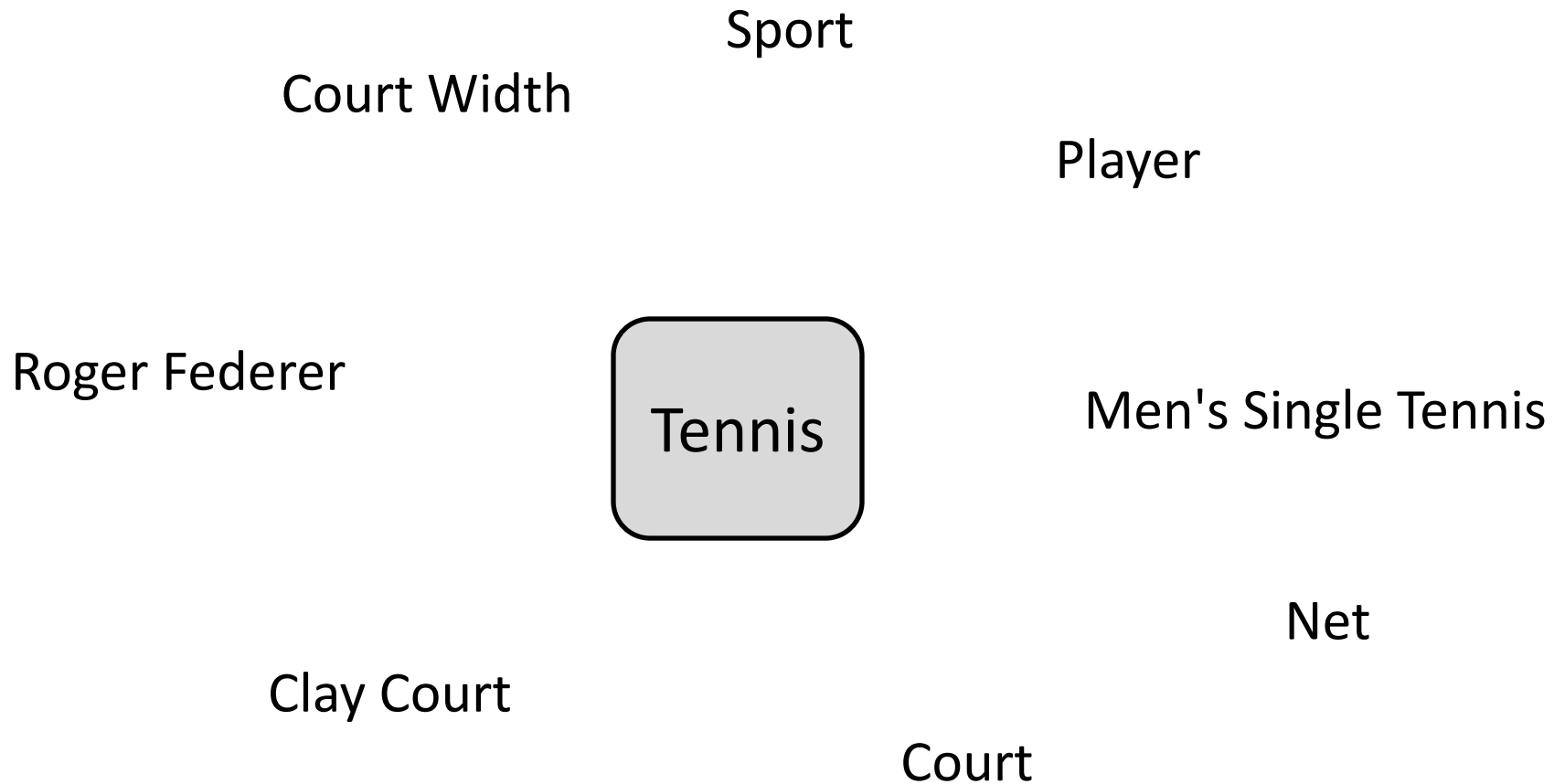
From the list of items below, create an ontology about plant by arranging them into a class hierarchy, as well as add relation (object property) and instance-of relations where appropriate.

(object property relation includes A has B, A has B as its member, A has B in it, A owns B.)





Exercise





Why ontologies are important? [2]

- Ontological analysis clarifies the structure of knowledge.
 - Given a domain, its ontology forms the heart of any system of knowledge representation for that domain.
- Ontologies enable knowledge sharing.
 - The ontology captures the intrinsic conceptual structure of the domain.
 - Suppose we perform an analysis and arrive at a satisfactory set of conceptualizations, and their representative terms
 - We can share the ontology with others who have similar needs for knowledge representation in that domain, thereby eliminating the need for replicating the knowledge-analysis process.
- Ontologies do not only introduce a sharable and reusable knowledge representation but can also add new knowledge about the domain.



Why do we use ontology?

- Why would someone want to develop an ontology? Some of the reasons are:
 - To share common understanding of the structure of information among people or software agents
 - To enable reuse of domain knowledge
 - To make domain assumptions explicit
 - To separate domain knowledge from the operational knowledge



Use of ontologies: As knowledge base for other programs

- Often an ontology of the domain is not a goal in itself.
- Developing an ontology is akin to defining a set of data and their structure for other programs to use.
 - Problem-solving methods, domain applications, and software agents use ontologies and knowledge bases built from ontologies as data.
 - For example: ontology of wine and food and appropriate combinations of wine with meals can be used as a basis for some applications in a suite of restaurant-managing tools:
 - Wine suggestion application for the menu of the day or answer queries from waiters and customers.
 - Application for recommending particular wines to purchase for upcoming menus or cookbooks



Use of Ontologies: For Better Data Management

- Some of the major characteristics of ontologies:
 - They make explicit domain assumptions.
 - They ensure a common understanding of information
- As a result, the interconnectedness and interoperability of the model make it invaluable for addressing the challenges of accessing and querying data in large organizations.



Example of Ontologies in real world

- Many disciplines now develop standardized ontologies that domain experts can use to share and annotate information in their fields.
- Medicine, for example, has produced large, standardized, structured vocabularies, for example:
 - SNOMED (Price and Spackman 2000)
 - The semantic network of the Unified Medical Language System (Humphreys and Lindberg 1993).
- Broad general-purpose ontologies are emerging as well, for example:
 - UNSPSC ontology which provides terminology for products and services (www.unspsc.org).



Use of Ontology in AI

- In AI, while knowledge representation pervades the entire field, two application areas in particular have depended on a rich body of knowledge.
 - Natural-language understanding
 - Domain knowledge often plays a crucial role in disambiguation.
 - Well-designed domain ontology provides the basis for domain knowledge representation
 - Domain ontology helps identify the semantic categories that are involved in understanding discourse in that domain
 - Knowledge-based problem solving, such as diagnosis, planning, and design
- Need commonsense knowledge and domain-specific knowledge



The Benefits of Using Ontologies

- By having the essential relationships between concepts built into them, they **enable automated reasoning about data**. Such reasoning is easy to implement in semantic graph databases that use ontologies as their semantic schemata.
- Ontologies provide **more coherent and easy navigation** as users move from one concept to another in the ontology structure.
- Ontologies are **easy to extend** as relationships and concept matching are easy to add to existing ontologies.
- Ontologies provide **the means to represent any data formats**, enabling smoother data integration, easier concept and text mining, and data-driven analytics.



Ontology Engineering

- Ontology requirement specification
- Development and Refinement
 - Knowledge elicitation and formalization
 - Produce a mature and application-oriented target ontology according to the specification
- Evaluation
 - Prove the usefulness of the developed ontology and the associated software environment
- Maintenance
 - Generally, things are constantly changing.
 - These changes must be reflected in the developed ontology, with the guarantee of coherence and compatible upgrade



Ontology Engineering [2]

- Reasoning
 - Ensure the quality of an ontology
 - Can be used in different phase of the ontology life cycle
 - During ontology design
 - Reasoning can be used to test whether concepts are non-contradictory, and to derive implied relations
 - During deployment
 - For determining the consistency of facts stated in annotations, or infer relationships between instances and ontology classes
 - Interoperability and integration of different ontologies
 - After asserting some inter-ontology relationships, the integrated class hierarchy is computed, and the concepts are checked for consistency.



GROUP EXERCISE: DESIGNING ONTOLOGY



Exercise

- 3-4 person per group
- Choose one topic from the list below
 - Clothes
 - Shoes
 - Printed Material
 - Vegetable
 - Accessories (เครื่องประดับ)
 - Pet
 - Hotel
 - Flower
 - Movies
 - Music/Song

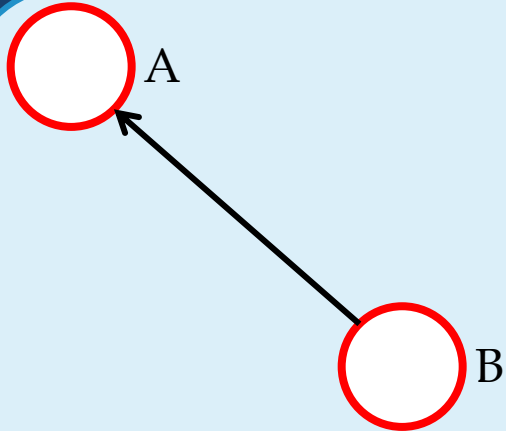


Exercise

For each concept, define:

1. Definition
2. Parent concept
3. Child concept
4. Sibling concepts
5. Property (both object and data property)

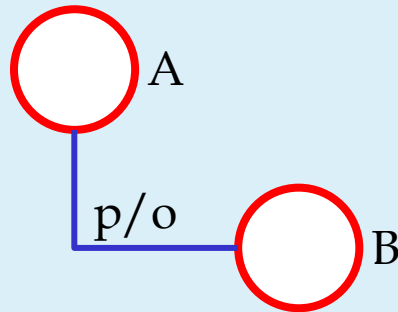
Notations



“B is-a A”

A (superclass)

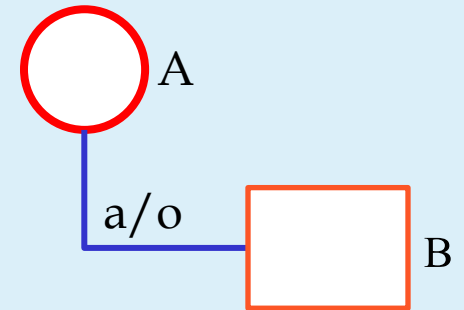
B (subclass)



“B related to class A”

A (core class)

B (constraint class)



“B is an attribute of A”

A (core class)

B (data type)

is-a = vertical relation within a tree

property = horizontal relation across trees

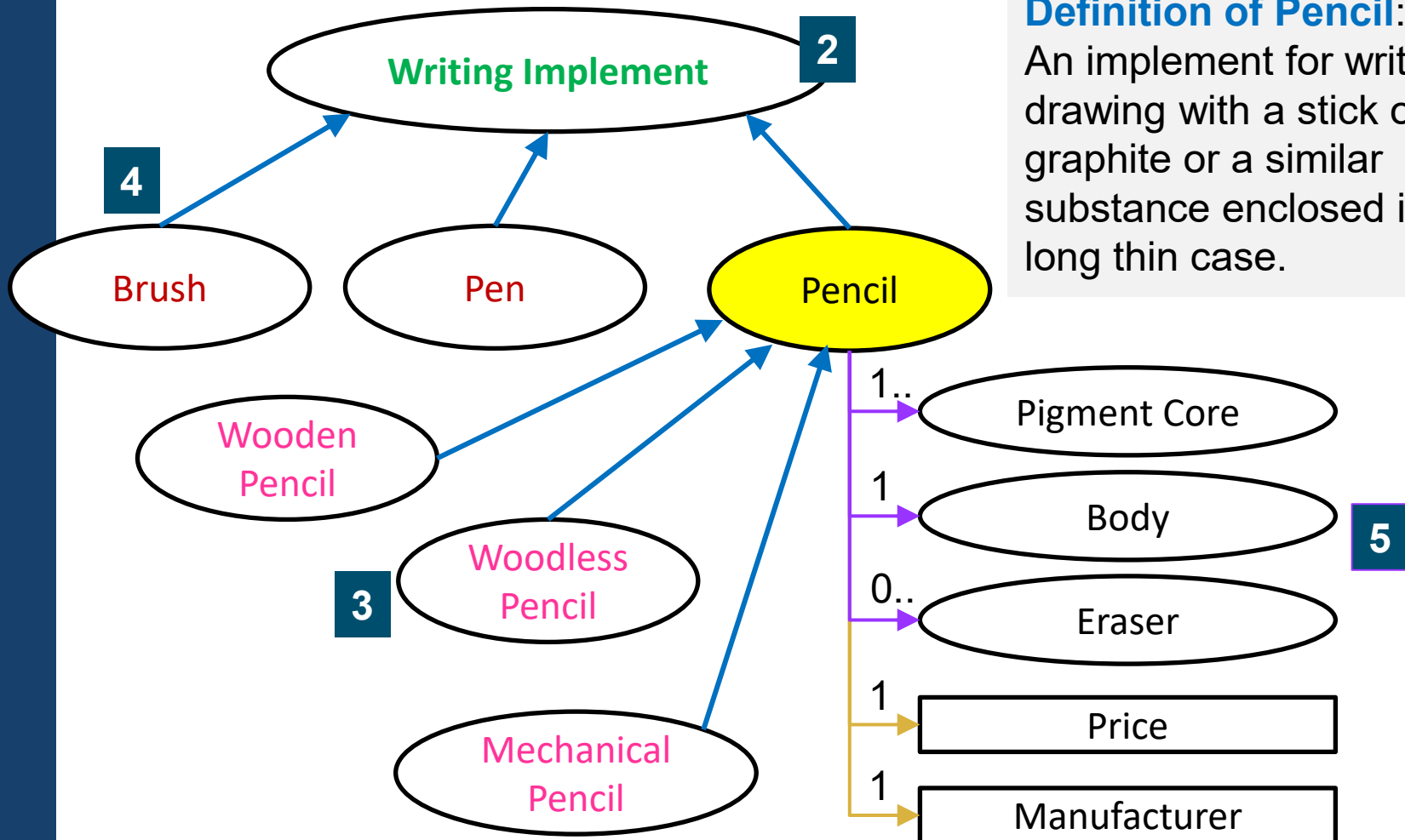


Example: PENCIL

1. Definition
2. Parent concept
3. Child concept
4. Sibling concept
5. Property

Definition of Pencil:

An implement for writing or drawing with a stick of graphite or a similar substance enclosed in a long thin case.





Useful knowledge sources for designing hierarchy and properties

- Google
- Thesaurus, WordNet, and other online dictionaries
 - Hyponym
 - Hypernym
 - Synonym
 - Etc.
- Wikipedia
- Etc.



Tips

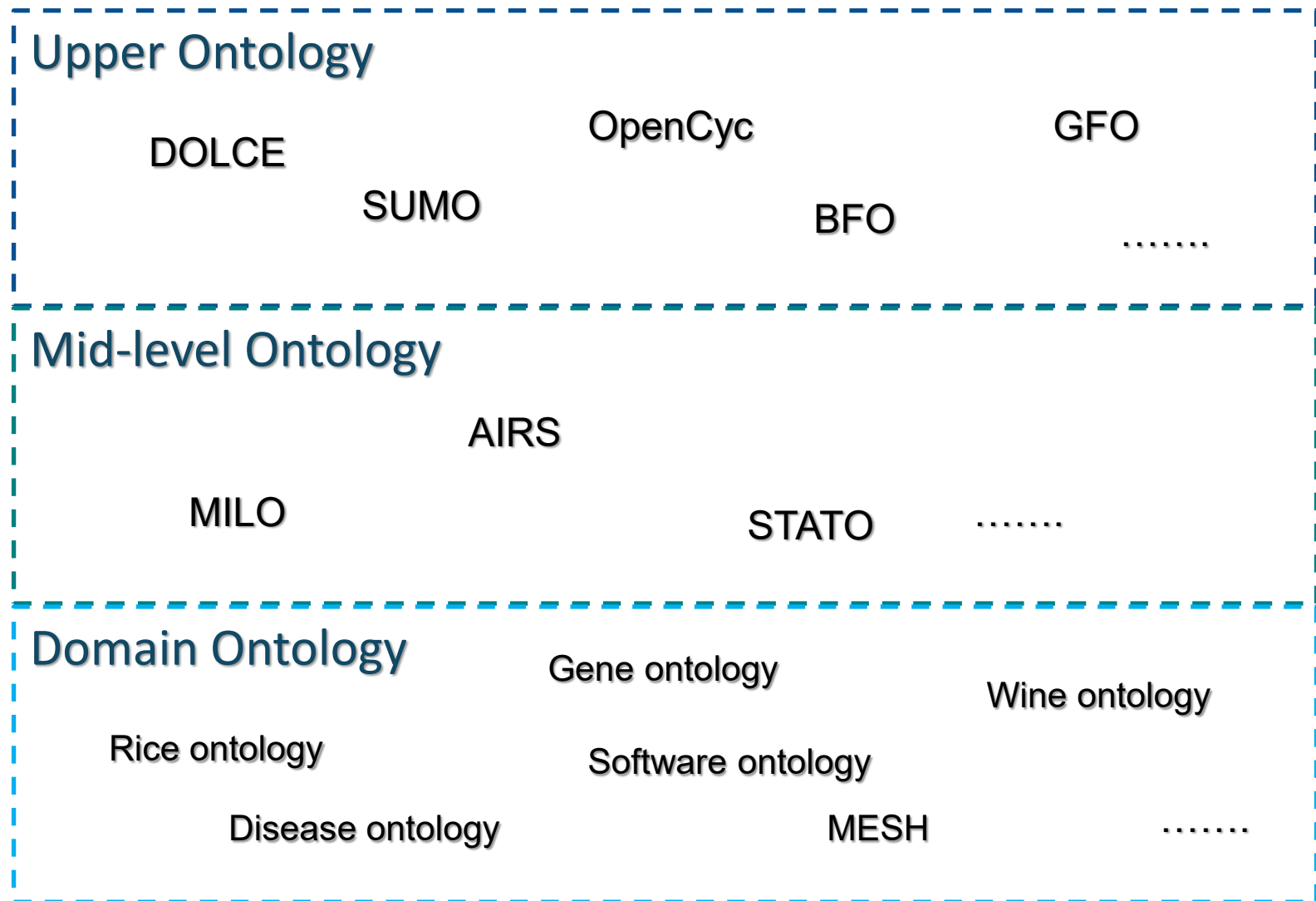
- Real knowledge can be very vast and difficult. Focus on the scope. List all possible concepts then remove them if irrelevant. Make sure you know the facts of it.
- IS-A relation is very strict: Remember it must inherit properties from its parent node.
 - When creating subclasses of a certain class, it is recommended to use the same criteria, with the same granularity.
- Object property should be a closed set of concepts. Otherwise, data property is more suitable.
- This knowledge engineering skill is not only for ontology development but also useful to organize your thought. Training is required to master the skill.



BROAD CLASSES OF ONTOLOGIES



Broad classes of Ontologies





Upper Ontology

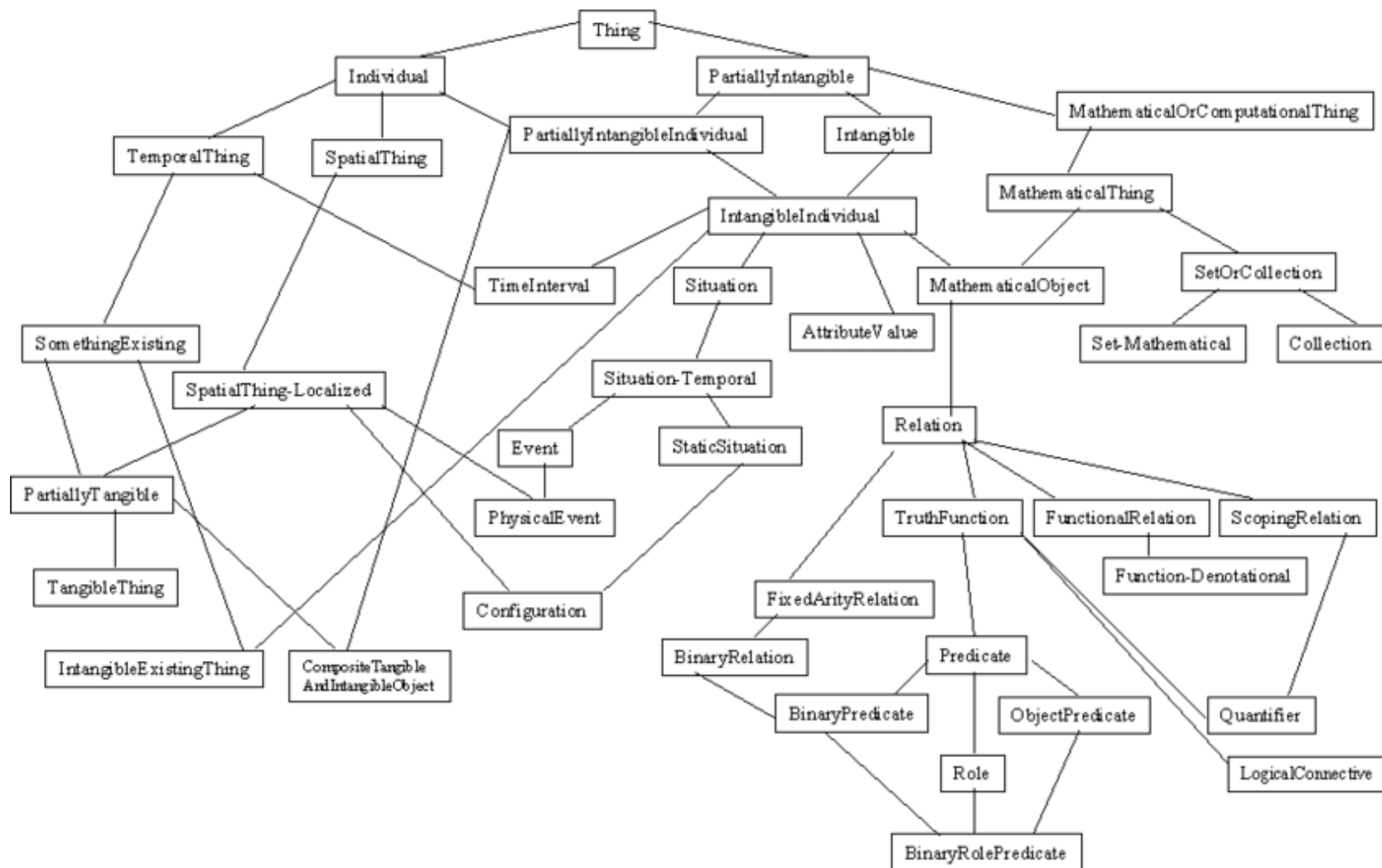
- Upper ontology or Foundational ontology or Universal ontology
 - Generic ontologies applicable to various domains
 - Define basic notions such as objects, relations, events, process, etc.
- An upper ontology is a high-level, domain-independent ontology
 - Provide a domain independent conceptual model that aims to be highly re-usable across specific domain applications
 - Often characterized as representing common sense concepts, i.e. those that are basic for human understanding of the world
- The concepts expressed in upper ontologies are intended to be basic and universal concepts
 - Usually describe very general level or abstract concepts
 - Ensure generality and expressivity for a wide area of domains.



Upper Ontology (cont.)

- Provide a framework by which disparate systems may utilize
 - One of the primary purposes of upper ontologies is to aid semantic integration across ontologies and to encourage a set of design principles within those ontologies that use them
- Can be used for deriving more domain-specific ontologies
- Domain or core reference ontologies based on the same foundational ontology can be more easily integrated
- More information on upper ontologies
 - A Comparison of Upper Ontologies
 - <http://www.disi.unige.it/person/MascardiV/Download/DISI-TR-06-21.pdf>

Example of OpenCyc





Mid-Level Ontology

- Serves as a bridge between abstract concepts defined in the upper ontology and low-level domain specific concepts specified in a domain ontology.
 - May provide more concrete representations of abstract concepts found in the upper ontology.
- Mid-level (as well as upper) ontologies are intended to provide a mechanism to make this mapping of concepts across domains easier.
- Represent knowledge at an intermediate level of detail independently of a specific task
- This ontology category also includes the set of ontologies that represent commonly used concepts, such as Time and Location.



Domain Ontology

- Specifies concepts particular to a domain of interest and represents those concepts and their relationships from a domain specific perspective.
- Domain ontology is only applicable to a domain with a specific view point.
 - View point defines how a group of users conceptualize and visualize some specific phenomenon
- Could be linked to a specific application



Domain Ontology

- Domain ontologies may be composed by importing mid-level ontologies. They may also extend concepts defined in mid-level or upper ontologies.
 - Reusing well-established ontologies in a domain ontology development allows one to take advantage of the semantic richness of the relevant concepts and logic already built into the reused ontology
 - Using common mid-level and upper ontologies can ease the process of integrating or mapping domain ontologies.
- Ontologies designed for specific tasks are called “*application ontologies*”



Local Ontologies/Application Ontologies

- Specialization of domain ontologies
 - There could be NO consensus or knowledge sharing
- Represents the particular model of a domain according to a single viewpoint of a user or a developer.



Reference

- Slides based on

[1] An Introduction to Ontologies and Ontology Engineering

- Catherine Roussey, Francois Pinet, Myoung Ah Kang, Oscar Corcho
- http://oa.upm.es/10381/1/An_Introduction.pdf

[2] What Are Ontologies and Why Do We Need Them

- B. Chandrasekaran, John R. Josephson, and V. Richard Benjamins

[3] W3C OWL guide

- <http://www.w3.org/TR/owl-guide/>

[4] Ontology Development 101: A Guide to Creating Your First Ontology

- Natalya F. Noy and Deborah L. McGuinness
- https://protege.stanford.edu/publications/ontology_development/ontology_101.pdf

[5] <http://www.slideshare.net/marinasantini1/09-semantic-webontologies>

[6] Representing and Managing Knowledge with Ontology: Ontology Engineering

- Dr. Taneth Ruangrajitpakorn